Serverless computing

WORKSHOP OVERVIEW: MULTI-USER URL SHORTENER

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Serverless

Serverless vs FaaS

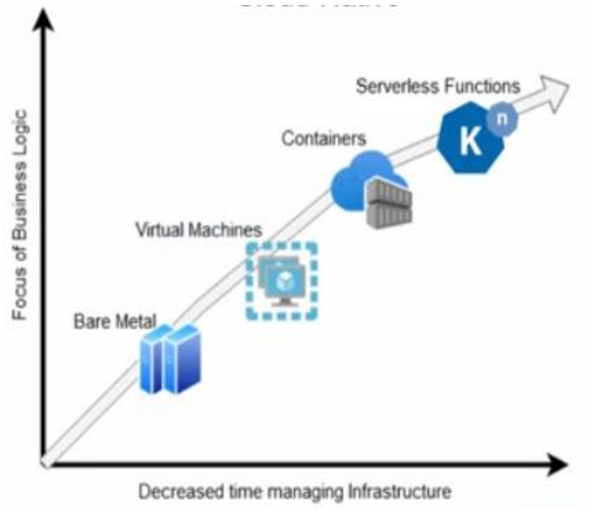
serverless Architecture

FaaS
- State / execution Time / latency …

Serverless platforms

Cloud execution model that enables
- a simpler,
- more cost-effective way

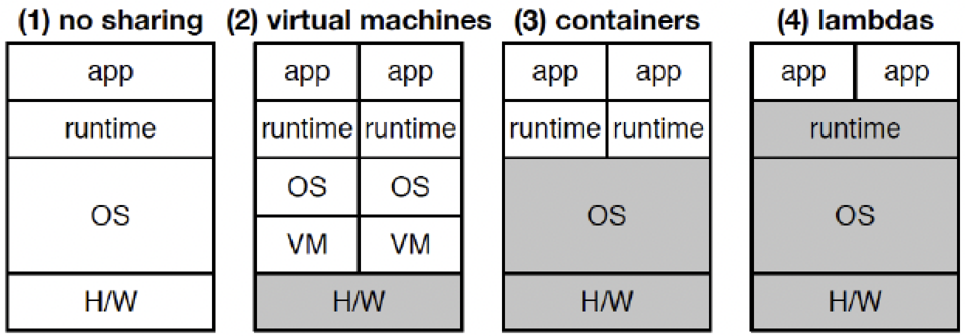
to build and operate cloud-native applications.



What is serverless?

Serverless is a cloud execution model that **automatically**

- **provisions** the computing resources
- **scales** resources up or down
- **scales** resources to zero when the application stops running.



Serverless

Serverless aims to
- eliminate the management
- and configuration tasks

enabling **users** to focus on the application.

The server is third-party “Backend as a Service” (BaaS)

BaaS could be:
- Serverless databases
- DevOps pipelines
- Kubernetes
- Etc.

Feature	Serverless (FaaS)	Backend-as-a-Service (BaaS)
Custom Logic	Write your own code (functions).	Use pre-built services (no code).
Use Case	Custom backend logic (e.g., image processing).	Pre-built features (e.g., auth, databases).
Control	Full control over function logic.	Limited to provider's offerings.
Scalability	Auto-scales per request.	Depends on the service (often auto-scaled).
Examples	AWS Lambda, Azure Functions.	Firebase, Auth0, Supabase.

Serverless vs. FaaS

FaaS is a subset of serverless –

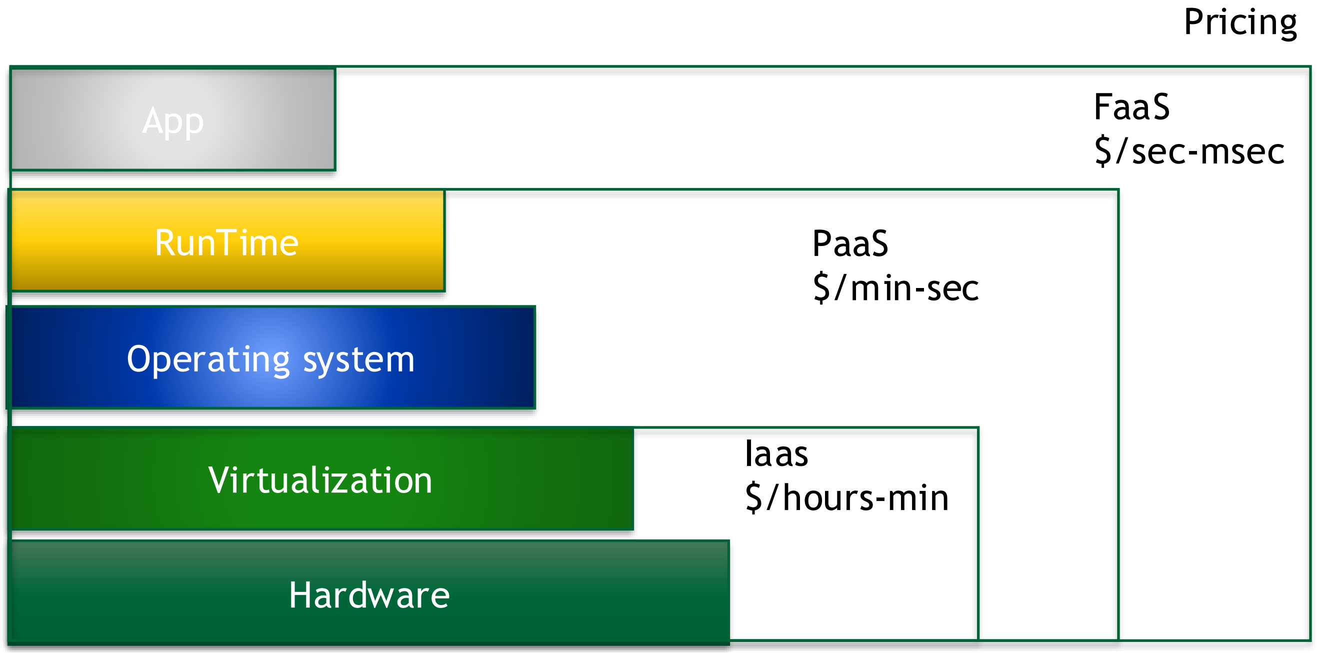
- compute paradigm
- application code run in **response to events.**

Serverless includes FaaS **+**

+ all the cloud services / resources supporting the Function
  - e.g. storage, databases, networks, API gateways, authentication

Key attributes of the Serverless approach

- Provisioning time
- No administrative burden of the server
- Elastic scaling
- Capacity planning
- Maintenance
- High availability and disaster recovery
- Resource utilization
- Charging granularity and billing



Serverless pros and cons

PROS

Enables developers to
- **focus** on writing code
- **Pay** only for execution Time
- **code** in **multiple languages**

Simplifies deployment and DevOps cycles

CONS

- **Cold start** (“higher” latency)
- Stable or predictable workloads
- Managing state is relatively complex
- Monitoring and debugging
- Vendor lock-in
  - Serverless architectures are designed to take advantage of an ecosystem of managed cloud services

Execution duration

FaaS functions are **limited in how long** each invocation is **allowed to run.**

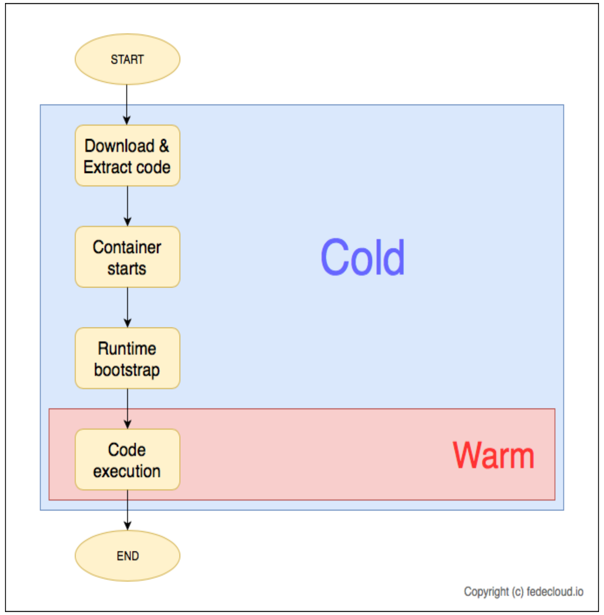
- the "**timeout**" for an AWS Lambda function to respond to an event is at most five minutes*, before being terminated.

Note: **long-lived tasks** are not suited to FaaS functions without re-architecture

- you need to create several different coordinated FaaS functions

Startup latency and "cold start"

It takes some time for a FaaS platform to initialize an instance of a function before each event.



Cold-start latency depends on many variables:

- the language,
- The number of libraries,
- The size the code,
- the configuration of the function environment itself,
- whether you need to connect to VPC resources, etc.
- The number of events processed
  - 10 events/second vs 1 event/hour - runtime boot of 50ms

When to use FaaS

Architectural Requirements	Design Red Flags	Neutral (Context-Dependent)
Statelessness	Synchronous Dependencies	Latency
Idempotency	Long Execution Time/function	Workload type
Algorithmic efficiency	Deployment artifact size	
Update Frequency	Vendor Locking	

Serverless platform

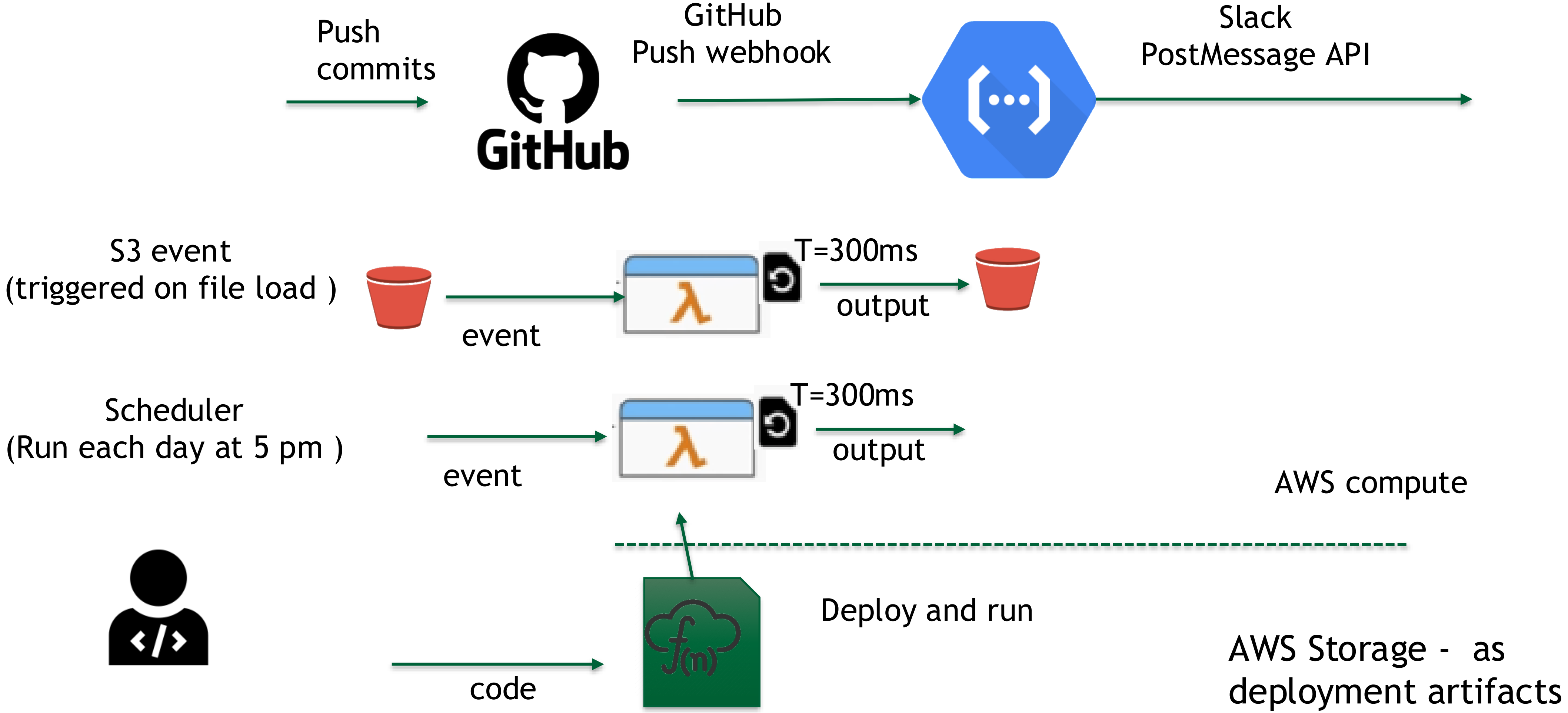
OPENSOURCE

Project	Maturity	Best Use Case
Knative*	Graduated (CNCF)	Enterprise-grade **K8s** serverless.
OpenFaaS*	High	Simple, portable functions for **K8s**.
OpenWhisk	Mature (Apache)	Large-scale distributed event-processing.
Fn Project	Active (Oracle)	Container-native serverless workflows.
Fission*	Niche	High-performance, fast-start functions for **K8s**
OpenLambda	Research	Exploring new serverless architectures (Wasm).

COMMERCIAL

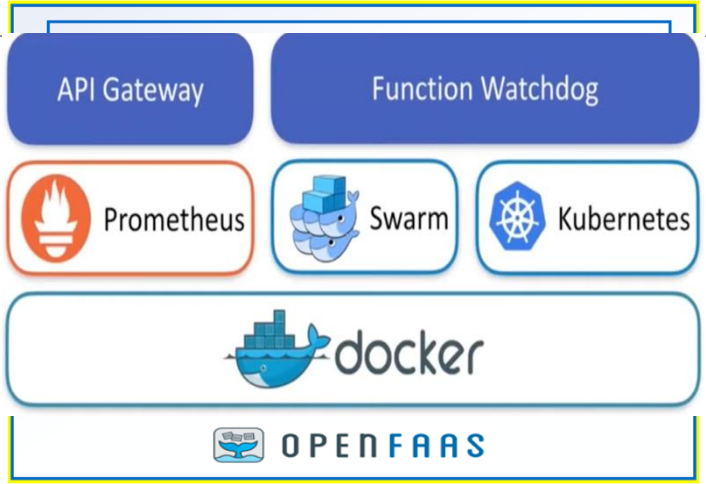
- AWS Lambda
- Google Cloud Functions
- Microsoft Azure
- Oracle Cloud Functions
- Alibaba Functions
- IBM Bluemix/OpenWhisk

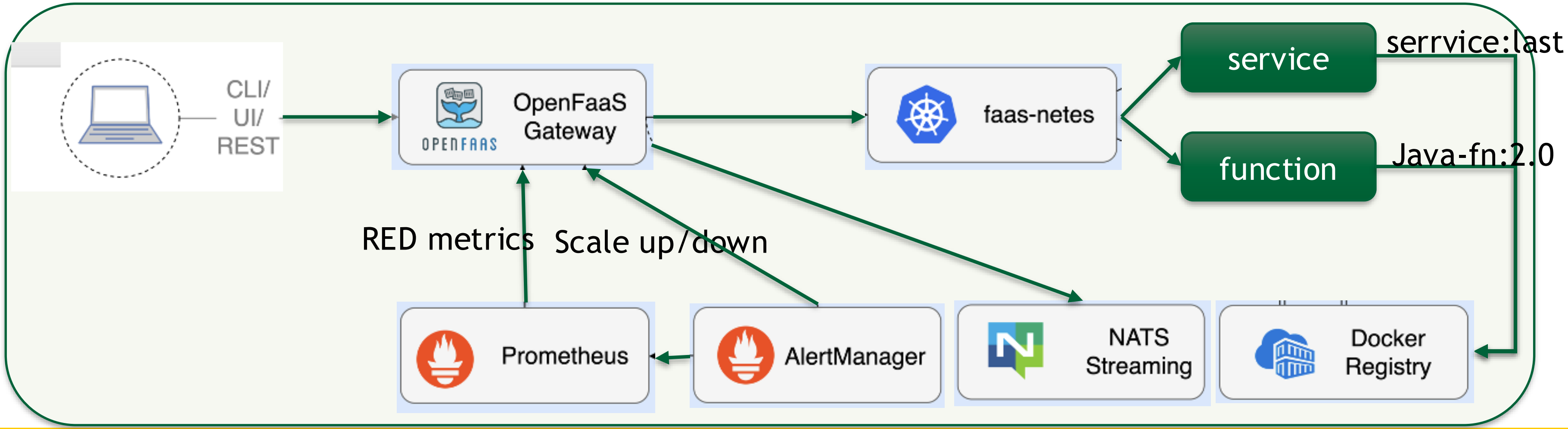
Functions in action



OpenFaas

- unit of execution for a **function** is a Pod.
- A **container registry** holds functions as **immutable artifacts** that can be deployed via the OpenFaaS gateway.





Source: OpenFaaS: From Zero to Serverless in 60 Seconds Anywhere with Alex Ellis https://www.youtube.com/watch?v=C3agSKv2s_w

On behalf of the DIGITAfrica consortium

Thank you!